Numerical Analysis I MAA 1203.
Chaper 1
Number System

The most used number system is decimal, Other well known number system are binary, octal, hexadecimal.

- Decimal (10)
  used digits 0, 1, 2, 3, 4, 5, 6, 7, 8, 9

- Binary (2)
  used digits 0, 1

- Octal (8)
  used digits 0, 1, 2, 3, 4, 5, 6, 7.

- Hexadecimal (16)
  used digits 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F

$$[d_n, \ldots, d_3\, d_2\, d_1]_\beta \leftarrow \text{Base}$$

$$\beta^{n-1}, \ldots, \beta^2 \quad \beta^1 \quad \beta^0$$

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A number

$$\xleftarrow{\hspace{3em}} \text{digits} \xrightarrow{\hspace{3em}}$$

$$[d_n, \dots, d_3\, d_2\, d_1 \,.\, d_1'\, d_2'\, d_3', \dots, d_n']_\beta$$

$$\beta^{n-1} \quad \beta^2 \quad \beta^1 \quad \beta^0 \quad \beta^{-1} \quad \beta^{-2} \quad \beta^{-3} \quad \beta^{-n}$$

$$\Downarrow$$

$$(d_n \times \beta^{n-1}) + \dots + (d_3 \times \beta^2) + (d_2 \times \beta^1) + (d_1 \times \beta^0) + (d_1' \times \beta^{-1}) + (d_2' \times \beta^{-2}) + (d_3 \times \beta^{-3}) + \dots + (d_n' \times \beta^{-n})$$

- Number of used digits $= \beta - 1$
- Numbers used $= 0 \leq N \leq \beta - 1$
- Base $= \beta$

Converting the Base

Ex : Decimal to Binary

Let x be positive decimal integer

$$x = (a_n\, a_{n-1}, \dots, a_1 a_0)$$

$$x = a_n 2^n + a_{n-1} 2^{n-1} + \dots + a_1 2^1 + a_0 2^0 = x_0$$

$$\frac{x_0}{2} = a_n 2^{n-1} + a_{n-1} 2^{n-2} + \dots + a_1 2^0 + \frac{a_0}{2} = x_1 + \frac{a^0}{2}$$

$$\frac{x_1}{2} = a_n 2^{n-2} + a_{n-1} 2^{n-3} + \dots + \frac{a_1}{2} = x_2 + \frac{a_1}{2}$$

$$\frac{x_2}{2} = a_n . 2^{n-3} + a_{n-1} 2^{n-4} + \dots + \frac{a_2}{2} = x_3 + \frac{a_2}{2}$$

$$\vdots$$

$$\frac{x_{n-2}}{2} = a_n . 2^1 + a_{n-1} + \frac{a_{n-2}}{2} = x_{n-1} + \frac{a_{n-2}}{2}$$

$$\frac{x_{n-1}}{2} = a_n + \frac{a_{n-1}}{2} = x_n + \frac{a_{n-1}}{2}$$

$$\frac{x_n}{2} = \frac{a_n}{2}.$$

This procedure is performed untill we get the zero quotient.

$(49.725)_{10}$ into other system

<u>To Binory</u>

2	49		0.725		0
2	24	– 1	0.725 × 2	= 1.45	0.1
2	12	– 0	0.45 × 2	= 0.90	0.10
2	6	– 0	0.9 × 2	= 1.8	0.101
2	3	– 0	0.8 × 2	= 1.6	0.1011
2	1	– 1	0.6 × 2	= 1.2	0.10111
	0	– 1	0.2 × 2	= 0.4	0.101110

110001.10111_2 //

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To octal

$$8\,|\,49$$
$$8\,|\,6 - 1$$
$$0 - 6$$

0.725	0
0.725 × 8 = 5.8	0.5
0.8 × 8 = 6.4	0.56
0.4 × 8 = 3.2	0.563
0.2 × 8 = 1.6	0.5631

$$61.5631_8 \ //$$

To hexadecimal

$$16\,|\,49$$
$$16\,|\,3 - 1$$
$$0 - 3$$

0.725	0
0.725 × 16 = 11.6	0.B
0.6 × 16 = 9.6	0.B9
0.6 × 16 = 9.6	0.B99

$$31.B99_{16} \ //$$

Binary to decimal

$$(1\ 1\ 0\ 0\ 0\ 1 . 1\ 0\ 1\ 1\ 1)_2$$
$$2^5\ 2^4\ 2^3\ 2^2\ 2^1\ 2^0 . 2^{-1}\ 2^{-2}\ 2^{-3}\ 2^{-4}\ 2^{-5}$$

$$(1 \times 32) + (1 \times 16) + (1 \times 1) + (1 \times \tfrac{1}{2}) + (1 \times \tfrac{1}{8}) + (1 \times \tfrac{1}{16}) + (1 \times \tfrac{1}{32})$$

$$49.725_{10} \ //$$

Octal to decimal

$$\begin{array}{cccccc} (6 & 1 . & 5 & 6 & 3 & 1)_8 \\ 8^1 & 8^0 . & 8^{-1} & 8^{-2} & 8^{-3} & 8^{-4} \end{array}$$

$$(6\times8)+(1\times1)+(5\times\frac{1}{8})+(6\times\frac{1}{64})+(3\times\frac{1}{512})+(1\times\frac{1}{4096})$$

$$(49.725)_{10}$$

Hexadecimal to decimal

$$\begin{array}{ccccc} (3 & 1 . & B & 9 & 9)_{16} \\ 16^1 & 16^0 & 16^{-1} & 16^{-2} & 16^{-3} \end{array}$$

$$(3\times16)+(1\times1)+(11\times\frac{1}{16})+(9\times\frac{1}{256})+(9\times\frac{1}{4096})$$

$$49.725_{10}$$ //

Binary to octal & hexadecimal

$$\begin{array}{c|ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 & 1 \\ 1 & 4 & 2 & 1 & 4 & 2 & 1 \\ 1 & & 5 & & 1 & & \end{array} \qquad \begin{array}{c} 2 \\ \downarrow \\ 8 \end{array}$$

$$151_8$$

$$\begin{array}{ccc|cccc} 1 & 1 & 0 & 1 & 0 & 0 & 1 \\ 4 & 2 & 1 & 8 & 4 & 2 & 1 \\ & & 6 & 9 & & & \end{array} \qquad \begin{array}{c} 2 \\ \downarrow \\ 16 \end{array}$$

$$69_{16}$$

Hexadecimal & octal to Binary

		1		5		1			
4	2	1	4	2	1	4	2	1	$8 \downarrow 2$
0	0	1	1	0	1	0	0	1	

$= 1101001_2$ //

		6		9					
8	4	2	1	8	4	2	1		$16 \downarrow 2$
0	1	1	0	1	0	0	1		

1101001_2 //

Floating point number system

A floating point number system is denoted by

$$F(\beta, k, m, M)$$

β = base
k = number of digits
m = minimum exponent
M = maximum exponent
e = exponent

it is expressed as

$$\pm(0.d_1 d_2 d_3, \dots, d_k)_\beta \times \beta^e$$ or

$$d_1 \cdot \beta^{-1} + d_2 \cdot \beta^{-2} + \dots + d_k \beta^{-k}$$

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Ex: $F(10, 2, 0, 2)$

$\pm(0.d_1d_2)_{10} \times 10^e$

$0 \leqslant e \leqslant 2$

$e = 0, 1, 2$

$1 \leqslant d_1 \leqslant 9$

$0 \leqslant d_2 \leqslant 9$

when $e = 0$

$0.d_1d_2 \times 10^0 = 0.d_1d_2$

$= 0.10, 0.11, \ldots, 0.19$

$0.20, 0.21, \ldots, 0.29$

$\vdots$

$0.90, 0.91, \ldots, 0.99$

when $e = 1$

$0.d_1d_2 \times 10^1 = d_1.d_2$

$= 1.0, 1.1, \ldots, 1.9$

$= 2.0, 2.1, \ldots, 2.9$

$\vdots$

$9.0, 9.1, \ldots, 9.9$

when $e=2$

$0.d_1d_2 \times 10^2 = d_1d_2$

$= 10, 11, \ldots, 19$

$20, 21, \ldots, 29$

$\vdots$

$90, 91, \ldots, 99$

100, 110, 120

... 190

200, 210

290

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Number of element in $F(10,2,0,2)$

$$|F(10,2,0,2)| = 541$$

$F(10,2,0,2) = \{0\} \cup \{0.d_1d_2 \times 10^0 \,/\, 1 \le d_1 \le 9,\ 0 \le d_2 \le 9\} \cup$
$\{0.d_1d_2 \times 10^1 \,/\, 1 \le d_1 \le 9,\ 0 \le d_2 \le 9\}$
$\cup \{0.d_1d_2 \times 10^2 \,/\, 1 \le d_1 \le 9,\ 0 \le d_2 \le\}$

Real numbers into floating point numbers.

Let y real number

$$y = \pm(0.d_1d_2,\ldots,d_kd_{k+1},\ldots)_\beta \times \beta^e$$

then floating point number of y with k number of digits.

$$fl(y) = \pm(0.d_1d_2,\ldots,d_k)_\beta \times \beta^e$$

There are two methods to do this.

1 chopping - Here, all the digits after d_k are ignored.

2. rounding — if $d_{k+1} < \beta/2$, write up to d_k
— if $d_{k+1} \geq \beta/2$, $0.d_1d_2,\ldots d_k + \frac{1}{\beta^k}$

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Ex: $\sqrt{7.1} = 2.66458$ to $F(10,2,0,2)$

1. chopping

$fl(\sqrt{7.1}) = 2.6$

6 > 5

~~7~~ > 6

2 rounding

3.0

$fl(\sqrt{7.1}) = 2.7 \Rightarrow 6$ 10/2

6 > 5

6 > 3

7

Rounding error

Absolute error $= |y - fl(y)|$

5

Relative error $= \dfrac{|y - fl(y)|}{|y|} \times 100\%$

— The machine precision (epsilon)

$$u \rightarrow \begin{cases} \beta^{1-k} & [\text{chopping}] \\ \frac{1}{2}\beta^{1-k} & [\text{rounding}] \end{cases}$$

$(2.25)_{10}$

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Floating point arithmatic

Let x and y be real numbers

$$x \oplus y = fl[fl(x) + fl(y)]$$

$$x \ominus y = fl[f(x) - fl(y)]$$

$$x \otimes y = fl[f(x) \times fl(y)]$$

$$x \oslash y = fl[f(x) \div fl(y)]$$

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Finite Differences.

Let X be finite different set of argument

$$X = \{x_1, x_2, x_3, \ldots, x_k, x_{k+1}, \ldots x_n\}$$

where $x_{k+1} - x_k = c$ (constant)

and each x value has corresponding y values.

$$Y = \{y_1, y_2, y_3, \ldots, y_k, y_{k+1}, \ldots, y_n\}.$$

$$\begin{array}{ccccccc} X & x_1 & x_2 & x_3 & x_k & x_{k+1} & x_n \\ \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\ Y & y_1 & y_2 & y_3 & y_k & y_{k+1} & y_n \end{array}$$

but every Δy is not equal.

Then $y_{k+1} - y_k$ called first difference and it is denoted by

$$\boxed{\Delta y_k = y_{k+1} - y_k}$$

ex: $\Delta y_1 = y_2 - y_1$
$\Delta y_2 = y_3 - y_2$
$\Delta y_3 = y_4 - y_3$

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The second difference

$$\Delta^2 y_k = \Delta(\Delta y_k)$$
$$= \Delta(y_{k+1} - y_k)$$
$$= \Delta y_{k+1} - \Delta y_k$$
$$= y_{k+2} - y_{k+1} - (y_{k+1} - y_k)$$
$$= y_{k+2} - 2y_{k+1} + y_k.$$

In generally

$$\boxed{\Delta^n y_k = \Delta^{n-1} y_{k+1} - \Delta^{n-1} y_k}$$

$$\Delta^n y_k = \Delta[\Delta^{n-1} y_k] = \Delta^{n-1}[\Delta y_k]$$

Ex:

x_k	1	3	5	7	9	11
y_k	3	7	10	15	16	21

$$\Delta y_2 = y_3 - y_2 = 10 - 7 = 3$$
$$\Delta y_3 = y_4 - y_3 = 15 - 10 = 5$$

$$\Delta^3 y_3 = \Delta^2 y_4 - \Delta^2 y_3$$
$$= \Delta y_5 - \Delta y_4 - (\Delta y_4 - \Delta y_3)$$
$$= y_6 - y_5 - (y_5 - y_4) - (y_5 - y_4 - (5))$$
$$= 21 - 16 - (16 - 15) - (16 - 15 - 5)$$
$$=$$

x_i	y_i	Δy_i	$\Delta^2 y_i$	$\Delta^3 y_i$
1	3			
		Δy_1		
3	7		$\Delta^2 y_1$	
		Δy_2		$\Delta^3 y_1$
5	10		$\Delta^2 y_2$	
		Δy_3		$\Delta^3 y_2$
7	15		$\Delta^2 y_3$	
		Δy_4		$\Delta^3 y_3$
9	16		$\Delta^2 y_4$	
		Δy_5		
11	21			

Ex. express $\Delta^3 y_0$ and $\Delta^4 y_0$ of y values.

$$\begin{aligned}
\Delta^3 y_0 &= \Delta^2(\Delta y_0) \\
&= \Delta^2(y_1 - y_0) \\
&= \Delta^2 y_1 - \Delta^2 y_0 \\
&= \Delta(\Delta y_1) - \Delta(\Delta y_0) \\
&= \Delta(y_2 - y_1) - \Delta(y_1 - y_0) \\
&= \Delta y_2 - \Delta y_1 - \Delta y_1 + \Delta y_0 \\
&= y_3 - y_2 - (y_2 - y_1) - (y_2 - y_1) + y_1 - y_0 \\
\Delta^3 y_0 &= y_3 - 3y_2 + 3y_1 - y_0
\end{aligned}$$

$$\begin{aligned}
\Delta^4 y_0 &= \Delta(\Delta^3 y_0) \\
&= \Delta(y_3 - 3y_2 + 3y_1 - y_0) \\
&= \Delta y_3 - 3\Delta y_2 + 3\Delta y_1 - \Delta y_0 \\
&= y_4 - y_3 - 3(y_3 - y_2) + 3(y_2 - y_1) - (y_1 - y_0) \\
&= y_4 - 4y_3 + 6y_2 - 4y_1 + y_0
\end{aligned}$$

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Theorem 1

$$\boxed{\Delta^k y_0 = \sum_{i=0}^{k} (-1)^i \begin{bmatrix} k \\ i \end{bmatrix} y_{k-i}}$$

Ex: $\Delta^4 y_0$

$$\Delta^4 y_0 = \sum_{i=0}^{4} (-1)^i \begin{bmatrix} 4 \\ i \end{bmatrix} y_{4-i}$$

$$= (-1)^0 \left[\frac{4!}{(4-0)!\,0!}\right] y_{4-0} + (-1)^1 \left[\frac{4!}{(4-1)!\,1!}\right] y_{4-}$$

$$+ (-1)^2 \left[\frac{4!}{(4-2)!\,2!}\right] y_{4-2} + (-1)^3 \left[\frac{4!}{(4-3)!\,3!}\right] y_{4-}$$

$$+ (-1)^4 \left[\frac{4!}{(4-3)!\,3!}\right] y_{4-3}$$

Theorem 2

(a) If y_k Constant $\Delta y_k = 0$

(b) $\Delta [c_1 y_k + c_2 z_k] = c_1 \Delta y_k + c_2 \Delta z_k$.

(c) $\Delta y_k z_k = y_{k+1} \Delta z_k + z_k \Delta y_k$

(d) $\Delta \left(\frac{y_k}{z_k}\right) = \frac{z_k \Delta y_k - y_k \Delta z_k}{z_k \cdot z_{k+1}}$

$$\boxed{\Delta^n c^k = (c-1)^n c^k}$$

Factorial Polynomial

For a positive integer n

$$k^{(n)} = k(k-1)(k-2)\ldots[k-(n-1)]$$

$$k^{(0)} = 1$$
$$k^{(1)} = k$$
$$k^{(2)} = k(k-1)$$
$$k^{(3)} = k(k-1)(k-2)$$

Theorom

$$\boxed{\Delta k^{(n)} = nk^{n-1}}$$

Ex: $\Delta^2 k^{(n)} = \Delta(\Delta k^{(n)})$
$= \Delta[n k^{n-1}]$
$= n[\Delta k^{n-1}]$
$= n[n-1]k^{n-2}$ //

Theorom

$$\Delta^{(n)} k^{(n)} = n!$$

$$\Delta^m k^{(n)} = 0 \quad \text{if} \quad m > n$$

Recursive Formula

$$\boxed{k^{(n+1)} = (k-n)k^{(n)}}$$

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$$\boxed{k^{(-n)} = \frac{1}{(k+1)(k+2)\cdots(k+n)}}$$

Stirling Numbers.

Consider the factorial polynomial

$$k^{(n)} = k(k-1)(k-2)\cdots[k-(n-1)]$$

Then

$$\boxed{k^{(n)} = S_1^{(n)}k + S_2^{(n)}k^2 + S_3^{(n)}k^3 + \cdots + S_n^{(n)}k^n}$$

$S_1^{(n)}, S_2^{(n)}, S_3^{(n)} \cdots S_n^{(n)}$ are Stirling num.

Recursive formula for Stirling Numbers

$$\boxed{S_i^{(n+1)} = S_{i-1}^{(n)} - n S_i^{(n)}}$$

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Finding Stirling Numbers.

$$k^{(n)} = S_1^{(n)} k + S_2^{(n)} k^2 + S_3^{(n)} k^3 + \dots + S_n^{(n)} k^n$$

<u>$n=1$</u>

$$k^{(1)} = S_1^{(1)} k$$
$$k = S_1^{(1)} k$$
$$\boxed{S_1^{(1)} = 1}$$

<u>$n=2$</u>

$$k^{(2)} = S_1^{(2)} k + S_2^{(2)} k^2$$
$$= S_1^{(1+1)} k + S_2^{(1+1)} k^2$$
$$= [S_0^1 - 1 \cdot S_1^1] k + [S_1^1 - 1 \cdot S_2^1] k^2$$
$$= [0 - 1 \cdot 1] k + [1 - 1 \cdot 0] k^2$$
$$\boxed{k^{(2)} = -k + k^2}$$

<u>$n=3$</u>

$$k^{(3)} = S_1^{(3)} k + S_2^{(3)} k^2 + S_3^{(3)} k^3$$
$$= S_1^{(2+1)} k + S_2^{(2+1)} k^2 + S_3^{(2+1)} k^3$$
$$= [S_0^2 - 2 S_1^2] k + [S_1^2 - 2 S_2^2] k^2 + [S_2^2 - 2 S_3^2] k^3$$
$$= [0 - 2 \times -1] k + [-1 - 2 \cdot 1] k^2 + [1 - 2 \cdot 0] k^3$$
$$k^{(3)} = 2k - 3k^2 + k^3 \;//$$

$$\boxed{S_0^{(n)} = 0 \; / \; S_m^{(n)} = 0 \text{ if } n < m}$$

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Theorem

Power of k can be expressed as combination of factorial polynomials.

$$\boxed{k^{n} = S_1^{(n)} k^{(1)} + S_2^{(n)} k^{(2)} + S_3^{(n)} k^{(3)} + \cdots + S_n^{(n)} k^{(n)}}$$

Recursive formula

$$\boxed{S_i^{(n+1)} = S_{i-1}^{(n)} + i\,S_i^{(n)}}$$

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Chapter 2

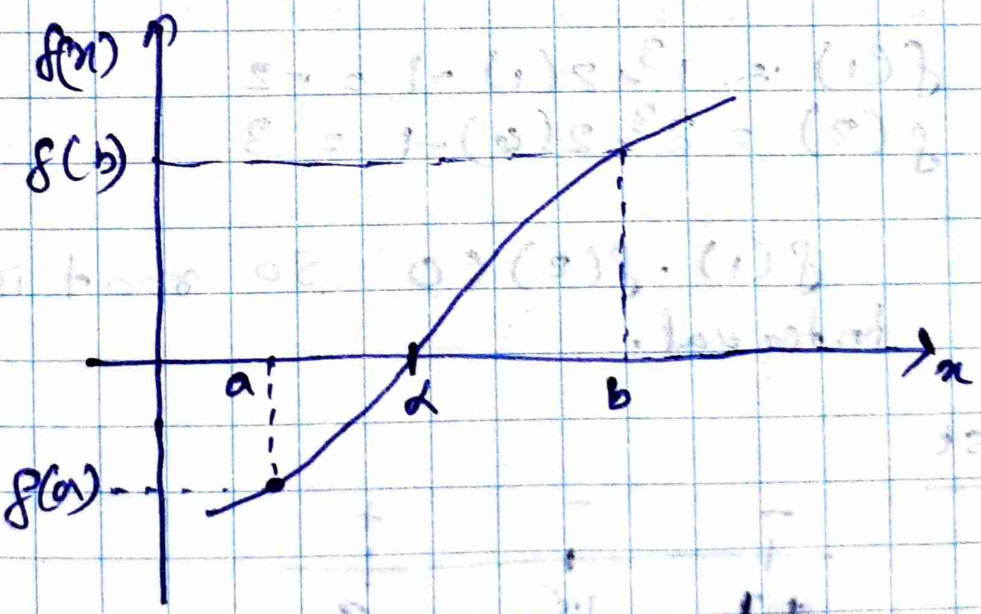
Finding the root of a function

The root of a function is a α s.t $f(\alpha)=0$ in a function $f(x)$

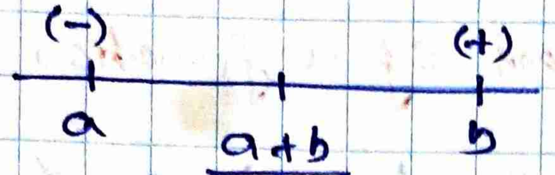
Bisection method.

Consider a function $f(x)$

If $f \in C[a,b]$ an $f(a)$ & $f(b)$ have opposit signs ($f(a)\cdot f(b)<0$), there exist $\alpha \in [a,b]$ such that $f(\alpha)=0$



Start bisection



1st

$$\frac{a+b}{2} = c$$

if $f(c) = (-)$, α is in $[c, b]$

if $f(c) = (+)$, α is in $[a, c]$.

Ex :- $f(x) = x^3 - 2x - 1$

$f(x) = 0$

$x^3 - 2x - 1 = 0$

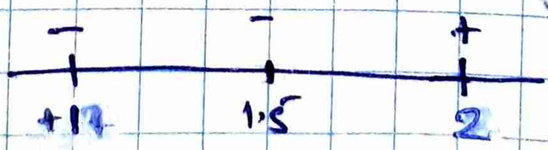
root (α) is in $(1, 2)$

$f(1) = 1^3 - 2(1) - 1 = -2$

$f(2) = 2^3 - 2(2) - 1 = 3$

$f(1) \cdot f(2) < 0$ so root is in above interval.

1st



$$\frac{1+2}{2} = \frac{3}{2} = 1.5$$

$f(1.5) = -0.625$

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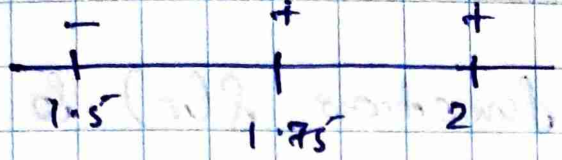
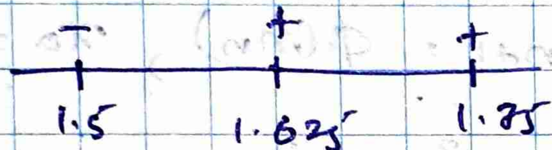
2nd



$$\frac{1.5+2}{2} = 1.75$$

$$f(1.75) = 0.859$$

3rd



$$\frac{1.5+1.75}{2} = 1.625$$

$$f(1.625) = 0.040$$

So on ....

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Iteration method.

Consider a function $f(x)$ & $f(x)=0$

- if the root is in interval (a, b)
- if $\phi(x)$ and $\phi'(x)$ is continously in (a,b) & $|\phi'(x)| < 1$, where $x = \phi(x)$
- Then $x_0, x_1, x_2, \ldots$ Converge the root.

where $x_{n+1} = \phi(x_n)$, $x_0 = a$

Ex: $f(x) = x^3 + 2x^2 + 10x - 20$

$f(x) = 0$

$x^3 + 2x^2 + 10x - 20 = 0$

$(1, 2)$

$f(1) = 1 + 2 + 10 - 20 = -27$

$f(2) = 8 + 8 + 20 - 20 = 16$

$f(1) \cdot f(2) < 0$ So root in $(1, 2)$

$x^3 + 2x^2 + 10x = 20$

$x(x^2 + 2x + 10) = 20$

$$x = \underbrace{\frac{20}{x^2 + 2x + 10}}_{\phi(x)}$$

$$\phi'(x) = \frac{-20(2x+2)}{x^2 + 2x + 10}$$

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$|\phi'(x)| < 1$ for all numbers in $(1, 2)$

So, Interation is valid.

$x_0 = 1$

$x_1 = \phi(x_0) = \dfrac{12 + 10 - 20}{1 + 2 + 10} = 1.5384..$

$x_2 = \phi(x_1) = \dfrac{20}{(1.5384)^2 + 2(1.5384) + 10} = 1.29$

So on.....

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Newton Raphson method.

Consider a function $f(x)$ & $f(x) = 0$.

- If root is in interval (a, b)
- $|g'(x)| < 1$ where

$$g(x) = x - \frac{f(x)}{f'(x)}$$

- Then $x_0, x_1, x_2, \ldots$ converge root where

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} \quad , x_0 = a$$

Ex: $f(x) = x^3 - 3x - 5$

$f(x) = 0$

$x^3 - 3x - 5 = 0$

$(2, 3)$

$f(2) = 8 - 6 - 5 = -3$

$f(3) = 27 - 9 - 5 = 13$

$f(2) \cdot f(3) < 0$ so root is in $(2, 3)$

$f'(x) = 3x^2 - 3$

$$g(x) = x - \frac{f(x)}{f'(x)}$$

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$$g(x) = x - \frac{x^3 - 3x - 5}{3x^2 - 3}$$

$$g(x) = \frac{2x^3 + 3x^2 - 3x + 5}{3x^2 - 3}$$

$|g'(x)| < 1$ (Show it)

$x_0 = 2$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = 2 - \frac{(2^3 - 3 \times 2 - 5)}{3 \times 2^2 - 3} = 2.33$$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} = 2 - \frac{(2.33^2 - 3 \times 2.33 - 5)}{3 \times 2.33^2 - 3}$$

$$= 2.280$$

So on.....

Regular falsi method.

Consider a function $f(x)$ & $f(x)=0$.

- If root is in interval (a, b)
- Then $x_0, x_1, x_2, \ldots$ converge root

where $x_{n+1} = x_n - f(x_n) \cdot \dfrac{(x_n - x_{n-1})}{f(x_n) - f(x_{n-1})}$

$x_0 = a \quad , x_1 = b.$

Ex:- $f(x) = x^3 - 2x - 5$

$f(x) = 0$

$x^3 - 2x - 5 = 0$

$(2, 3)$

$f(2) = 2^3 - 2(2) - 5 = -1$

$f(3) = 3^3 - 2(3) - 5 = 16$

$f(2) \cdot f(3) < 1$ So root is in $(2, 3)$

$x_0 = 2 \quad x_1 = 3$

$x_2 = x_1 - f(x_1) \dfrac{(x_1 - x_0)}{f(x_1) - f(x_0)}$

$= 3 - f(3) \dfrac{(3 - 2)}{f(3) - f(2)}$

$=$

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$$= 3 - \frac{16}{16 - (-1)}$$

$$x_2 = 2.0588$$

$$x_3 = x_2 - f(x_2)\frac{(x_2 - x_1)}{f(x_2) - f(x_1)}$$

$$= 2.0588 - f(2.0588)\frac{(2.0588 - 3)}{f(2.0588) - f(3)}$$

So on.....

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Chapter 3

Interpolation.

One of the most useful groups of functions used to interpolate an unknown function is the group of polynomials

Finding a polynomial ~~interpolation~~ $p(x)$ for which

$$p(x_k) = y_k = f(x_k)$$

Direct method

$$f(x) \approx \text{or} = p(x) = y$$

$x = x_0$

$$a_0 + a_1 x_0 + a_2 x_0^2 + \dots + a_n x_0^n = y_0$$
$$a_0 + a_1 x_1 + a_2 x_1^2 + \dots + a_n x_1^n = y_1$$
$$a_0 + a_1 x_2 + a_2 x_2^2 + \dots + a_n x_2^n = y_2$$
$$\vdots$$
$$a_0 + a_1 x_n + a_2 x_n + \dots + a_n x_n = y_n$$

$$\begin{bmatrix} 1 & x_0 & x_0^2 & \dots & x_0^n \\ 1 & x_1 & x_1^2 & \dots & x_0^n \\ \vdots & & & & \\ 1 & x_n & x_n^2 & \dots & x_n^n \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ \vdots \\ a_n \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

Lagrange Basis & Interpolation Formula

$$l_i(x) = \frac{(x-x_0)(x-x_1)\dots(x-x_{i-1})(x-x_{i+1})\dots(x-x_n)}{(x_i-x_0)(x_i-x_1)\dots(x_i-x_{i-1})(x_i-x_{i-1})\dots(x_i-x_n)}$$

$$p(x) = \sum_{i=1}^{n} f_i \, l_i(x)$$

$$y = f(x) \approx \text{ or } = p(x)$$

Eg:-

x_i	0.1	0.4	0.8
y_i	0.1003	0.4225	1.0296

$y_i = f(x) \approx p(x)$ here, $p(x) \Rightarrow p_2(x)$

$$p_2(x) = 0.1003\frac{(x-0.4)(x-0.8)}{(0.1-0.4)(0.1-0.8)} + 0.4225\frac{(x-0.8)(x-0.1)}{(0.4-0.8)(0.4-0.1)} + 1.0296\frac{(x-0.1)(x-0.4)}{(0.8-0.1)(0.8-0.4)}$$

Error of Interpolation

$$E_n(x) = f(x) - p_n(x)$$

$$E_n(x) \le \frac{1}{(n+1)!} \max_{a \le x \le b} \left| (x-x_0)(x-x_1)\cdots(x-x_n) \right|$$

Consider $f(x) \in C[a,b]$

$$E_n(x) = f(x) - p_n(x)$$

$$E_n(x) \le \frac{1}{(n+1)!} \max_{a \le x \le b} \left| (x-x_0)(x-x_1)\cdots(x-x_n) \right| M_{n+1}$$

where, $M_{n+1} = \max_{a \le x \le b} \left| f^{n+1}(x) \right|$

Eg:- Consider $(x_0, f_0), (x_1, f_1), (x_2, f_2)$

$x_1 = x_0 + h$ =, $x_0 = -h$

$x_2 = x_0 + 2h$.

Show that the error bounded in value by $\frac{h^3 M_3}{9\sqrt{3}}$ where $M_3 = \max|f'''(x)|$, $x \in [a,b]$

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$$E_n(x) = \frac{1}{(n+1)!} \left| \max_{a \le x \le b} (x-x_0)(x-x_1)\dots(x-x_n) \right| M_{n+1}$$

$$M_{n+1} = \max_{a \le x \le b} f^{n+1}(x)$$

$$E_2(x) = \frac{1}{(2+1)!} \left| \max_{[x_0, x_2]} (x-x_0)(x-x_1)(x-x_2) \right| M_{2+1}$$

$$M_{2+1} = \max_{[x_0, x_2]} f'''(x)$$

$$E_2(x) = \frac{1}{6} \left| \max_{[x_0, x_2]} g(x) \right| M_3$$

$$g(x) = (x+h)(x-h+h)(x+h-2h)$$
$$g(x) = x^3 - xh^2$$

$$g'(x) = 3x^2 - h^2 = 0$$
$$x = \pm \frac{h}{\sqrt{3}}$$

$$g\left(\pm\frac{h}{\sqrt{3}}\right) = 3\left(\frac{h}{\sqrt{3}}\right)^3 - h^2 = \frac{2h^3}{3\sqrt{3}}$$

$$E_2(x) \le \frac{1}{6 \times 3\sqrt{3}} \times 2h^3 M_3 = \frac{h^3 M_3}{9\sqrt{3}}$$

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Collocation polynomial

$$f(x) = p_n(x) = f(x_0) + \frac{f'(x_0)}{1!}(x - x_0) + \frac{f''(x_0)(x - x_0)^2}{2!} + \dots \frac{f^n(x_0)(x - x_0)^n}{n!}$$

$$E_n(x) = f(x) - p_n(x)$$

$$= \frac{f^{n+1}(\alpha)(x - x_0)^{n+1}}{(n+1)!}$$

See the eg in note.

- grid / mesh points

$$x_0, x_1, x_2, \dots, x_N$$

$$x_n = x_0 + nh$$

$$h = \frac{b-a}{N}$$

$x_0 \quad x_1 \quad x_2, \dots$ $\qquad y_n$ — approximate

$y_0 \quad y_1 \quad y_2, \dots$ $\qquad y(x_n)$ — exact

y_{n+1} at x_{n+1} is obtained using related data only at x_n

Euler

$$\left.\begin{aligned} &\bullet\ \frac{dy}{dx} = f(x,y) \\ &\bullet\ y_0 = y(a) \\ &\bullet\ y_{n+1} = y_n + h\,f(x_n, y_n) \end{aligned}\right\} \text{forward}$$

$$\left.\begin{aligned} &\bullet\ \frac{dy}{dx} = f(x,y) \\ &\bullet\ y_0 = y(a) \\ &\bullet\ y_{n+1} = x_n + h\,f(x_{n+1}, y_{n+1}) \end{aligned}\right\} \text{backward.}$$

Runge kutta

- $\frac{dy}{dx} = f(x,y)$
- $y_0 = y(a)$
- $y_{n+1} = y_n + \frac{h}{6}\left[k_1 + 2k_2 + 2k_3 + k_4\right]$

$$k_1 = f(x_n, y_n)$$

$$k_2 = f\left[x_n + \frac{h}{2}, y_n + \frac{h}{2}k_1\right]$$

$$k_3 = f\left[x_n + \frac{h}{2}, y_n + \frac{h}{2}k_2\right]$$

$$k_4 = f\left[x_n + h, y_n + hk_3\right].$$